

Day 19 - Lecture 19.12: Relational Plots in Seaborn (relplot, scatterplot, lineplot)

1. Overview & Purpose

Relational Plots are designed to visualize the **relationship between two or more continuous numerical variables**. Seaborn provides two primary relational plot types:

1. **Scatter Plot (`sns.scatterplot`)**: Ideal when each row represents an independent observation. Shows correlation, clusters, and spread.
 2. **Line Plot (`sns.lineplot`)**: Ideal when the x -axis represents a continuous, ordered progression (e.g. time, year, sequence). Automatically performs **aggregation and confidence interval estimation** for multiple measurements at each x .
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2. Automatic Aggregation & Confidence Intervals

When multiple records exist for a single x value (e.g., monthly passenger records across multiple years):

- `sns.lineplot()` computes the **mean** line: $\bar{y}_x = \frac{1}{m} \sum_{j=1}^m y_{\{x, j\}}$
- It draws a **shaded band** around the line representing the **95% Confidence Interval (CI)** (by default using bootstrapping) or Standard Deviation:

```
sns.lineplot(data=df, x='year', y='passengers', errorbar='ci') # 95% CI
sns.lineplot(data=df, x='year', y='passengers', errorbar='sd') # Standard De
sns.lineplot(data=df, x='year', y='passengers', errorbar=None) # No band
```

3. Code Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Scatter Plot on tips dataset
tips = sns.load_dataset("tips")

plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=tips,
    x="total_bill",
    y="tip",
    hue="time",
    palette="Dark2",
    s=70,
    alpha=0.85
)
plt.title("Scatter Plot: Total Bill vs Tip (Colored by Lunch / Dinner)", fontsize=12)
plt.xlabel("Total Bill ($)")
plt.ylabel("Tip ($)")
plt.show()

# 2. Line Plot with Aggregation on flights dataset
flights = sns.load_dataset("flights")

plt.figure(figsize=(9, 5))
sns.lineplot(
    data=flights,
    x="year",
    y="passengers",
    color="navy",
    marker="o",
    errorbar="sd" # Shows standard deviation across months
)
plt.title("Annual Flight Passengers Trend (Aggregated Across Months with SD)", fontsize=12)
plt.xlabel("Year")
plt.ylabel("Number of Passengers")
plt.grid(True, linestyle="--", alpha=0.5)
plt.show()
```

4. Key Takeaways & Best Practices

- Use `sns.scatterplot` to spot bivariate correlation and non-linear patterns.
- Use `sns.lineplot` whenever x represents time series or sequential tracking.
- The shaded area in `lineplot` is not an error in your code—it represents the statistical variance across sub-samples!